

Mechanics and energetics of medusan jet propulsion

THOMAS L. DANIEL¹

Department of Zoology, Duke University, Durham, NC, U.S.A. 27706

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A theoretical model is developed to describe the mechanics and energetics of medusae which swim by jet propulsion. The model leads to four generalizations: (i) The acceleration reaction, not drag, is the dominant fluid force experienced by swimming medusae. (ii) There exist behaviors such as contraction time and relative duration of the relaxation of contraction phases which maximize the velocity and efficiency and minimize the cost of locomotion during normal swimming. Also, escape performance is maximized with certain behaviors. Existing data for medusae show that many operate near these optima. (iii) The ratio of efficiency to cost of locomotion is a reasonable measure of swimming behavior and provides a means to study the relationships between swimming behavior and size. (iv) For medusae of a particular shape exhibiting a particular swimming behavior there exists a unique size which maximizes the ratio of efficiency to cost of locomotion.

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On trouvera ici la description d'un modèle théorique illustrant les propriétés mécaniques et énergétiques de méduses qui se déplacent par propulsion. Le modèle permet quatre généralisations: (i) une réaction d'accélération, pas de traînée, est la force liquide dominante chez les méduses en déplacement; (ii) certains comportements, tels la durée d'une contraction et les durées relatives des phases de relaxation et de contraction, minimisent le coût de la locomotion au cours de la nage normale, tout en maximisant la vitesse et l'efficacité. De plus, la performance de la nage au cours de la fuite est maximisée par certains comportements. Les données de la littérature démontrent que plusieurs méduses se servent de ces facteurs d'optimisation. (iii) Le rapport coût: efficacité de la locomotion constitue une bonne mesure du comportement de nage et permet d'étudier les relations entre le comportement de nage et la taille: (iv) chez les méduses de forme particulière qui ont un comportement de nage particulier aussi, une seule taille maximise le rapport efficacité: coût de la locomotion.

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Introduction

Animals use a variety of mechanisms to swim. Some, such as eels, sharks, and snakes, undulate their bodies to produce thrust; others, such as ostraciiform fishes, aquatic beetles, penguins, and sea turtles, swim by use of oscillating appendages; still others, such as squid and jellyfish, use a jet reaction mechanism to produce thrust. In all modes of swimming, animals expend energy to move various portions of their bodies and thereby to do work on their environment. And, for each mode, there must exist mechanical and energetic constraints on an animal's size, shape, and swimming behavior. In this paper, I take a theoretical view of these constraints for jet-propelled medusae.

Both the forces generated by an animal's swimming movements and the forces resisting its motion through the environment must be known in order to explore any constraints on an animal's morphology and swimming behavior. And, to determine these constraints from present theory, we are limited to animals of simple shape undergoing simple swimming movements. Complex swimming motions generate complex flow patterns for which analyses of resistive forces such as drag are difficult. While the mechanism of thrust production by

undulating fishes is well understood, we know little about the resistive forces these animals experience (Lighthill 1975; Wu 1977). By contrast, animals which swim with oscillating appendages, in general, have rigid bodies, and thus resistive forces may be easy to estimate. But the thrust produced by large amplitude oscillation of appendages is poorly understood (for an exception, see Katz and Weihs 1978; Blake 1981a, 1981b).

Jet-propelled animals such as squid and hydrozoan medusae have simple shapes and swimming movements; they therefore allow simple theoretical estimates of both propulsive and resistive forces. Previous studies of jet propulsion have emphasized (i) the origin of the propulsive force (Johnson *et al.* 1973; Weihs 1977) and (ii) the kinematics of swimming (Packard 1969; Gladfelter 1972, 1973; Donaldson *et al.* 1980), but they have neglected the resistive forces experienced by these animals. Here, a theoretical model is developed to determine both the resistive and propulsive forces experienced by swimming medusae. Analyses of these forces are used, in turn, to explore the mechanics and energetics of swimming in terms of a medusa's size, shape, and swimming behavior.

Medusan swimming movements consist of repeated contractions of an umbrella-shaped structure, the bell (Fig. 1). During each contraction, water is forcibly

¹Present address: California Institute of Technology, 104–44 Pasadena, CA, U.S.A. 91125.

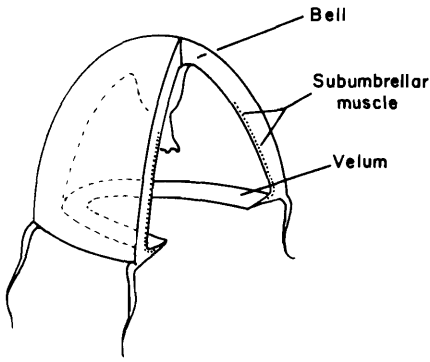


FIG. 1. Diagram of a medusa showing the location of the circularly oriented subumbrellar swimming muscles and the velum, a ring of tissue at the base of the bell. The dome-shaped portion of the animal is called the "bell."

expelled to provide the thrust which propels the animal by a jet reaction mechanism. Because thrust is produced in pulses, medusae swim in an unsteady manner, they are constantly accelerating and decelerating as they move through their environment. Because medusae move in a fluid, a force resists their motion. Part of this force is drag. Drag resists the motion of an animal in a fluid and depends on its instantaneous velocity relative to the fluid (for an overview see Vogel 1981). Associated with unsteady motion is an additional force called the acceleration reaction which resists *changes* in the animal's velocity (Batchelor 1967; Lighthill 1975; Blake 1979, 1981c; Webb 1979, 1982). In this paper, the model is developed to illustrate the role of the acceleration reaction in the mechanics and energetics of medusan swimming.

Energetic constraints on medusan swimming are explored using two familiar concepts: the "cost" and "efficiency" of locomotion (Schmidt-Nielsen 1977; Tucker 1975). Previous studies of animal swimming have provided theoretical and empirical estimates of locomotor efficiency (Lighthill 1975; Wu 1971; Webb 1971a, 1971b, 1975; Weihs 1973) and empirical estimates of the cost of locomotion (Brett 1964, 1965). Theoretical estimates of both parameters are used to ask: Does selection favor medusae which move about with the least cost or the greatest efficiency? Or does selection favor those animals which exhibit some compromise between cost and efficiency?

Swimming movements: behavior

The swimming movements of medusae are divided into two phases: contraction and relaxation. During contraction, circularly oriented subumbrellar muscles pull the sides of the bell inward, reducing the volume of the subumbrellar cavity, and forcing water out through the velar aperture (see Fig. 1). During the relaxation phase, water is drawn back into the subumbrellar cavity

as the bell returns to its resting shape. While bell contraction is a direct result of contraction of subumbrellar muscles, bell relaxation is not. There are no muscles which oppose the subumbrellar muscles (Gladfelter 1972, 1973). Instead, bell relaxation results from the release of elastic energy stored in the bell during the propulsive phase, just as a rubber band snaps back upon release of a load.

Medusae swim primarily to feed and to escape from predation (Mackie 1974). While feeding, they may swim with bouts of five or more repeated bell contractions followed by a period of inactivity (Leonard 1980; Mills 1981). This behavior will be called normal swimming. In contrast, escape swimming consists of only one bell contraction and the resultant relaxation phase. Both swimming behaviors may be characterized by a contraction duration, t_c , and a relaxation duration, t_r . The relative duration of these phases is given by:

$$[1] \quad \delta = t_r/t_c$$

A swimming cycle, then, consists of one contraction and one relaxation with a frequency of $1/(t_c + t_r)$.

The model: mechanics

The motion of medusae is modeled with a momentum balance for an accelerating animal. The model consists of four main terms, each of which corresponds to an instantaneous force produced by, or acting on the animal, thus

$$[2] \quad \text{Thrust} = \text{drag} + \text{acceleration reaction} \\ + \text{force to overcome inertia of animal}$$

This verbal equation may be written as a differential equation whose solution describes the kinematics of swimming medusae. The behavior of each of these terms and the assumptions underlying its mathematical formulation are discussed below. Definitions for all symbols are given in the Appendix as well as in the text.

Thrust

During the contraction phase, water is forcibly ejected from the subumbrellar cavity through the velar aperture providing thrust. The magnitude of this force is given by the rate of momentum efflux:

$$[3] \quad T = u_e(dm/dr)$$

where T is the thrust, u_e is the velocity of ejected fluid, and m is the instantaneous mass of the animal plus any water held in the subumbrellar cavity. The mass efflux rate is assumed to be constant within any given phase of a swimming cycle. The effects of this assumption are considered in the Discussion section under Mechanics.

Thrust acts in a direction opposite to the motion of the ejected fluid. Hence during the contraction phase, when water is expelled in an oral direction (away from the

apex of the bell), thrust acts in an aboral direction. Conversely during the relaxation phase, thrust acts in an oral direction, retarding the motion of the animal as water is pulled back into the bell. Thus a similar thrust term, but with opposite sign can be written for the relaxation phase.

The mass flow rate and the velocity of ejected fluid can be expressed in terms of the velar aperture area (A_v) and the rate of volume change of the animal during either phase (dV/dt):

$$[4] \quad u_e = (1/\rho A_v)(dm/dt)$$

$$[5] \quad dm/dt = \rho(dV/dt)$$

where ρ is the density of water. Substituting Eqs. 4 and 5 into Eq. 3 gives an expression for thrust with factors which can be determined experimentally from cinefilm analyses:

$$[6] \quad T = (\rho/A_v)(dV/dt)^2$$

Note that for a constant rate of volume change, the magnitude of the thrust is inversely proportional to the square of the duration of the contraction or relaxation phase:

$$[7] \quad T = (\rho/A_v)(\Delta V/\tau)^2$$

where τ is the duration of the phase. Outpocketing of the velum during the contraction phase tends to maintain a constant velar aperture area (see for example, Gladfelter 1972, pp. 60–61).

Drag

The instantaneous drag (D) acting on a swimming medusa is proportional to its dimensions and instantaneous velocity (u):

$$[8] \quad D = C_d 0.5 \rho S u^2$$

where S is the area of the animal projected in its direction of motion. The proportionality constant, C_d , in Eq. 8 is called the drag coefficient. As long as the animal moves relative to its fluid environment and regardless of the phase of the swimming cycle, drag resists its motion.

All factors in Eq. 8, except the density of water, vary in time for a swimming medusa. Besides the obvious variation in velocity, shape changes of the swimming animal make C_d and S variable in time. I assume that the rate of change of S is a simple function of the rate of volume change:

$$[9] \quad dS/dt = (3/2h)(dV/dt)$$

where h is the height of the bell. Equation 9 when solved with the assumption of a constant rate of volume change gives an expression for the instantaneous projected area of the animal:

$$[10] \quad S = S_0 + (3/2h)(dV/dt)t$$

where S_0 is the projected area of the animal at the beginning of the contraction phase and t is any instant in time. During the contraction phase, dV/dt is negative and S decreases. Conversely, the area increases during the relaxation phase.

The time dependence of the drag coefficient depends on its relationship with another dimensionless term, the Reynolds number (Re). The Reynolds number represents the ratio of inertial to viscous forces in a fluid moving steadily past some object (Vogel 1981):

$$[11] \quad Re = ul/\nu$$

where ν is the kinematic viscosity of the fluid and l is some characteristic length for the object. The choice of l is arbitrary; for spheres and cylinders, l is commonly taken as their diameter. Similarly, I use the instantaneous diameter of a medusae as its characteristic length. The drag coefficient is approximated from existing data for the drag coefficient of steadily moving spheres as a function of their Reynolds number (Hoerner 1965):

$$[12] \quad C_d = 24/Re^n$$

The power n is 1.0 for Reynolds numbers less than 1. The value of n is 0.7 for Reynolds numbers between 1 and 500. According to Gladfelter's (1973) data, medusae rarely exceed a Reynolds number of 500. Equations 10 through 12 in combination with Eq. 8 yield a time-dependent function for the drag acting on a swimming medusa in terms of its velocity and dimensions.

Acceleration reaction and inertia of animal

The acceleration reaction is given by Batchelor (1967) as:

$$[13] \quad G = -\alpha \rho V du/dt$$

where α is the added mass coefficient, a dimensionless term which reflects the effect of shape on the acceleration of fluid around an object (Fig. 2): prolate (cigar-shaped) objects have low added-mass coefficients; oblate objects have high added-mass coefficients. Size dependence of the acceleration reaction is reflected in the term for the animal's volume (V): larger medusae, by virtue of their larger volume require a greater force for a given acceleration. Drag, by contrast, is approximately proportional to an animal's projected area.

The total force (F) required to accelerate the animal (neglecting drag) is the sum of the acceleration reaction plus the force required to accelerate the mass of the animal itself:

$$[14] \quad F = (1 + \alpha)\rho V du/dt$$

Swimming movements give rise to volume and shape

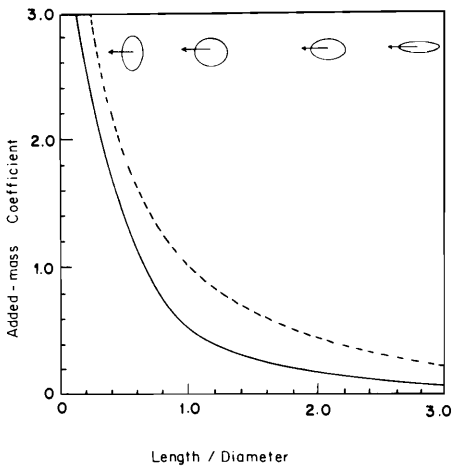


FIG. 2. A plot of the theoretical added-mass coefficient against the length to diameter (fineness) ratio for ellipsoids (—) and elliptical cylinders (---). The length to diameter ratio is the ratio of the axis parallel to the direction of motion to the axis perpendicular to the direction of motion. The axis of rotation for the ellipsoids is the length axis. (Values from Batchelor 1967; Kochin *et al.* 1964; Lamb 1932.)

changes of the bell throughout a single swimming cycle. Hence, Eq. 14 contains two time-varying parameters: the animal's volume and its added mass coefficient. Both parameters are written as time-dependent functions. The instantaneous volume of a medusa is given as:

$$[15] \quad V(t) = V_0 + (dV/dt)t$$

where V_0 is the volume of the animal at the beginning of the contraction phase. The rate of volume change, assumed to be constant, is expressed in terms of the instantaneous height and radius of the bell using standard mensuration formulas for hemiellipsoids. The added-mass coefficient is determined from the instantaneous bell height (h) and radius (r) with a regression equation for experimentally determined added-mass coefficients of hemiellipsoids (T. L. Daniel, in preparation):

$$[16] \quad \alpha = (h/r)^{1.4}$$

Balance of forces

Equation 2 can now be written as a differential equation:

$$[17] \quad (\rho/A_v)(dV/dt) = 0.5\rho SC_d u^2 + (1 + \alpha)\rho V(du/dt)$$

where C_d , u , α , and V are all time-dependent functions specified above. The term on the left-hand side of the equation represents the thrust force. Drag is given by the first term on the right-hand side of the equation. The acceleration reaction and the inertia of the animal are

given by the second term. This equation represents the balance of the instantaneous forces acting on a swimming animal. A fourth-order Runge-Kutta method (Golomb and Shanks 1965) is used for a numerical solution to Eq. 17.

The terms in Eq. 17 are used to define the ratio of the average acceleration reaction to the average total resistive force:

$$[18] \quad R = \frac{\int_0^\tau [\alpha\rho V(du/dt)]dt}{\int_0^\tau \{[0.5\rho SC_d u^2] + [\alpha\rho V(du/dt)]\}dt}$$

where R is the ratio. A value of 1.0 for R indicates no drag, 0 no acceleration reaction, and 0.5 equal contributions of drag and the acceleration reaction to the total force.

The model: energetics

The chief factors in the energetics of medusan swimming are the cost and efficiency of their locomotion. Cost of locomotion is defined as (Gabrielli and von Kármán 1950; Tucker 1975):

$$[19] \quad C = P_i/Wu$$

where P_i is the rate at which a moving animal consumes energy, W is its weight, and u is its velocity. W is the animal's mass times a constant ($g = 9.8 \text{ m} \cdot \text{s}^{-2}$), making cost a dimensionless term. This definition of cost applies to steady motion. Medusae, however, swim unsteadily and their mass varies with time. Thus the average mass and velocity of the animal are used to define the cost of locomotion for medusae.

The total energy expended is given by the power input:

$$[20] \quad E = \int_0^\tau P_i dt$$

The efficiency of locomotion is defined as the ratio of the "power out" (P_o , the rate at which work is done to the environment) to the "power in" (P_i , the rate of energy consumption (Tucker 1975)):

$$[21] \quad \eta = P_o/P_i$$

Both cost and efficiency are typically viewed in terms of continuous motion (Tucker 1975; Schmidt-Nielsen 1977). In escape swimming, though, minimization of cost and maximization of efficiency may not be the prime consideration; achieving a rapid acceleration and placing as large a distance as possible between prey and predator is critical to the survival of the escapee. Hence an appropriate "strategy" is to maximize the product of acceleration and distance for a successful escape. But associated with escape is a cost which is the energy required to produce the escape movements. An

appropriate measure of the ratio of cost to benefit is

$$[22] \quad \epsilon = E/(m \cdot a \cdot d)$$

where E is the energy required to escape at an average acceleration, a , through a distance, d , for a medusa of mass, m . Here, ϵ is the "escape index." A low value for the escape index is "better" than a high one.

Power input

A swimming medusa expends energy to produce the jet for thrust, to deform its bell, and to carry on basic physiological activities such as respiration. The latter is assumed to be negligible. The power required to produce a jet ($P_{i,t}$) is approximated by the time averaged integral of the product of the thrust and jet velocity:

$$[23] \quad P_{i,t} = (1/t_c) \int_0^{t_c} T u_e dt$$

where the integral is taken over the duration of the contraction phase.

During the relaxation phase, some of the energy imparted to the bell is used to suck water back in. The bell is made primarily of mesoglea which is, like most biological soft materials, viscoelastic (Chapman 1953; Alexander 1962; Gosline 1971). Some of the energy imparted to the bell is lost by the "viscosity" of the material (ultimately as heat) and upon removal of a load, some of the energy is returned by "elastic recoil." The total energy imparted to the bell (E_t) is equal to the energy required to fill the bell (E_f) plus the energy dissipated by the action of viscosity (E_v).

$$[24] \quad E_t = E_f + \Delta E_t$$

where Δ is the dissipation factor (Nielsen 1962). Rearranging Eq. 24 gives an expression for the total energy imparted to the bell in terms of the energy required to fill the bell:

$$[25] \quad E_t = E_f/(1 - \Delta)$$

The term $(1 - \Delta)$ is the efficiency of elastic energy storage. When the dissipation factor is 1.0, all of the energy imparted to the bell is lost by the action of viscosity and the efficiency of elastic energy storage is 0. When there is no viscous loss of energy in the bell, Δ is zero and all of the energy imparted to the bell is returned by elastic recoil.

The average rate at which energy is used to fill the bell ($P_{i,f}$) is given implicitly by Eq. 25:

$$[26] \quad P_{i,f} = [1/(1 - \Delta)](E_f/t_r)$$

E_f is given by an equation similar to Eq. 23:

$$[27a] \quad E_f = t_r P_{i,f}$$

$$[27b] \quad P_{i,f} = [1/t_r(1 - \Delta)] \int_0^{t_r} T u_e dt$$

where T and u_e are the thrust and velocity of the fluid entering the bell during relaxation. Equations 23 and 27 are combined to give an expression for the power input of swimming medusae:

$$[28] \quad P_i = P_{i,f} + P_{i,t}$$

The total energy required to produce a single swimming cycle is

$$[29] \quad E = \int_0^{t_r} P_{i,f} dt + \int_0^{t_c} P_{i,t} dt$$

Power output

Swimming medusae do work to overcome drag and the acceleration reaction and also do work to accelerate their own mass. The average rate at which work is done to overcome drag over some time interval τ is

$$[30] \quad P_{o,d} = 1/\tau \int_0^{\tau} 0.5\rho S C_d U^3 dt$$

To accelerate their own mass and overcome the acceleration reaction, medusae do work at an average rate of:

$$[31] \quad P_{o,a} = 1/\tau \int_0^{\tau} (1 + \alpha)\rho V (dU/dt) U dt$$

The total average power output is the sum of these two components.

Model parameters

A "standard medusa" is defined here to illustrate the parameters necessary for the model. Values for these parameters bracket those presented by Gladfelter (1972, 1973) for a variety of hydrozoan medusae.

(i) Dimensions of the bell. The standard medusa has a bell height, h , and a bell radius, r , of 1.0 cm. The velar aperture area is $\frac{1}{2}$ the area of the base of the bell.

(ii) Phase durations. The duration of the contraction phase (t_c) is 0.1 s and the relative phase duration (δ) is 2.0 for the standard medusa.

(iii) Volume change. Volume of the subumbrellar space decreases by 50% during the contraction phase and increases by the same amount during the relaxation phase.

(iv) Physical properties of the fluid. At 20°C, the density and kinematic viscosity of water are 1.0 g/cm³ and 0.01 cm²/s, respectively.

Simulation results

The standard medusa: performance

Figure 3 shows the predicted kinematics of the standard medusa in terms of the distance traveled per swimming cycle (Fig. 3a), velocity (Fig. 3b) and acceleration (Fig. 3c). Two important features of

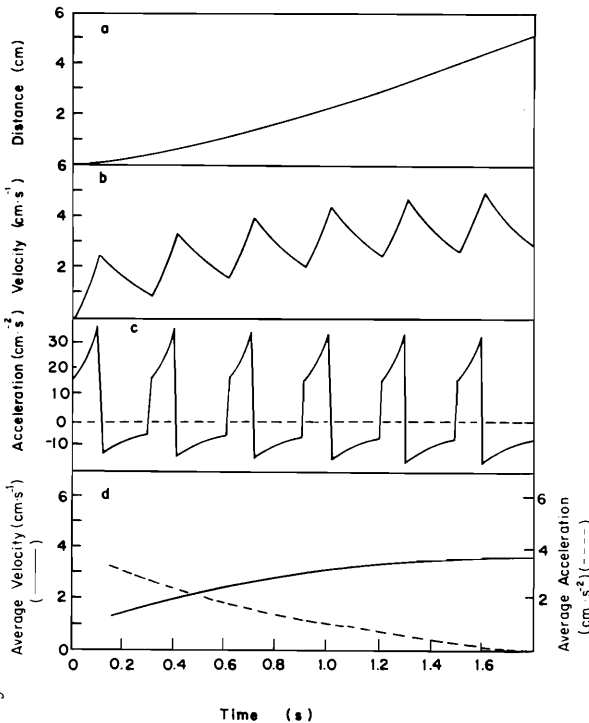


FIG. 3. The predicted swimming kinematics of the standard medusa (bell height = 1 cm, bell radius = 1 cm, $t_c = 0.1$ s, $\delta = 2$) undergoing six consecutive swimming cycles after starting from rest. (a) Distance traveled per swimming cycle. (b) Instantaneous velocity. (c) Instantaneous acceleration are plotted against time. (d) A plot of the average velocity (—) and the average acceleration (---) as a function of time. Note that velocity is not uniform in time, the motion is highly unsteady.

medusan locomotion are evident: (i) The velocity and acceleration of a swimming medusa are not uniform in time: the motion is highly unsteady. (ii) Both the velocity and acceleration averaged over one complete swimming cycle also change in time (Fig. 3d). After four or five consecutive swimming cycles the average velocity approaches some asymptotic maximum value and the average acceleration decays towards zero.

The instantaneous forces resisting the motion of the standard medusa are summarized in Fig. 4 for the first and fifth swimming cycle after starting from rest. These results show that the acceleration reaction is the dominant fluid force. The importance of the acceleration reaction decays the longer a medusa swims after starting from rest. This result follows from the increase in the average velocity (hence, an increase in drag) of a medusa undergoing consecutive swimming cycles and the decrease in acceleration noted previously. But, as an instantaneous force, the acceleration reaction constitutes more than 50% of the total fluid force even after many consecutive swimming cycles (Fig. 4, broken line). Notice also that the sign of the acceleration reaction changes between the contraction and relaxation

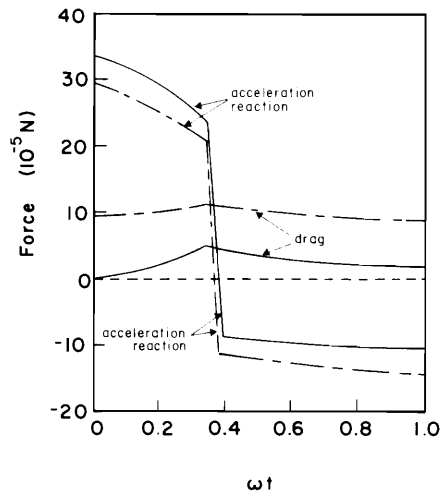


FIG. 4. Predicted instantaneous resistive forces acting on the standard medusa are plotted against the fraction of the duration of a swimming cycle (ωt) for the first (—) and fifth (---) cycle after starting from rest. Those lines which remain positive represent drag; the other lines represent the acceleration reaction.

phases. During the contraction phase the acceleration reaction is positive, resisting the forward acceleration of the animal. During the relaxation phase, the acceleration reaction is negative, resisting deceleration and thus tending to keep the animal moving. Figure 5 shows that neglecting the acceleration reaction yields an overestimate of animal's swimming velocity for the first few swimming cycles but has little effect after about five consecutive cycles.

The relative contribution of the acceleration reaction averaged over one complete swimming cycle is considerably less than its maximum instantaneous contribution. Figure 6 summarizes this result for the standard medusa with a plot of the relative contribution of drag (line a), the inertia of the medusa (line b), and

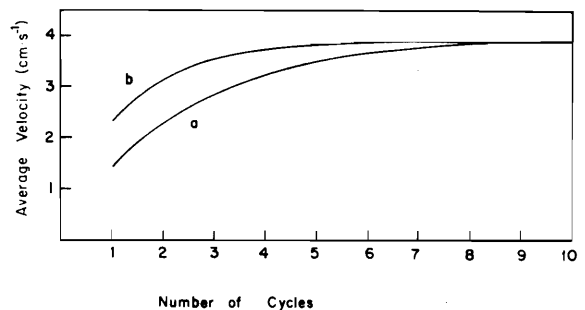


FIG. 5. The predicted average velocity of the standard medusa is plotted against the number of swimming cycles after starting from rest. The lower line (a) represents simulations which include the acceleration reaction. The upper line (b) corresponds to simulations which neglect the acceleration reaction.

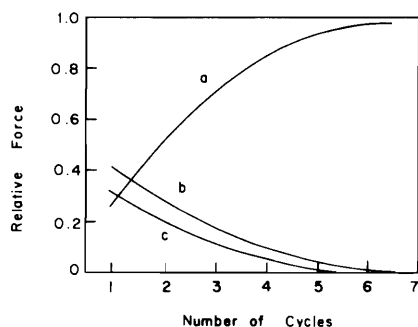


FIG. 6. The relative contribution of the acceleration reaction (*c*), the inertia of the animal (*b*), and drag (*a*) to the total force is plotted against the number of swimming cycles after starting from rest for the standard medusa.

the acceleration reaction (line *c*). Notice that in the first cycle after starting from rest the acceleration reaction is greater than drag. Hence, in escape swimming (one cycle to escape) it dominates the fluid forces resisting a medusa's motion. After the animal swims for several consecutive swimming cycles, drag dominates the average force resisting its motion. Thus, depending on the type of behavior, we expect qualitatively different relative magnitudes of each force: as an instantaneous force and for escape swimming, the acceleration reaction is most important. Drag is the dominant average force during continuous swimming.

Both the cost and efficiency of medusan locomotion depend strongly on forces related to the acceleration of the animal. As a medusa swims for longer times, the cost of locomotion decreases and the efficiency increases even though the average drag force increases. Figure 7 summarizes this result for the standard medusa.

The efficiency of medusan locomotion also depends on the efficiency with which the bell stores elastic energy. Figure 8 shows this relationship with a plot of

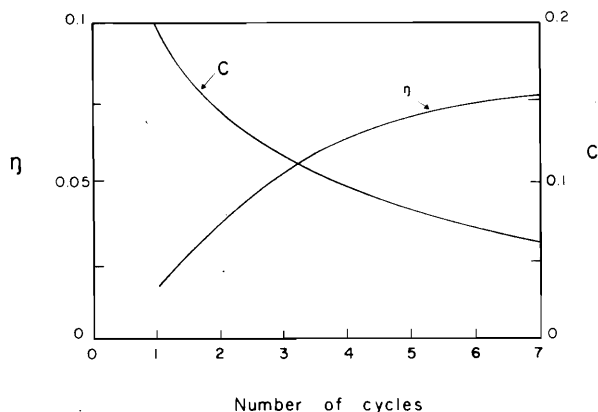


FIG. 7. The predicted cost (*C*) and efficiency (η) of locomotion are plotted against the number of swimming cycles after starting from rest for the standard medusa.

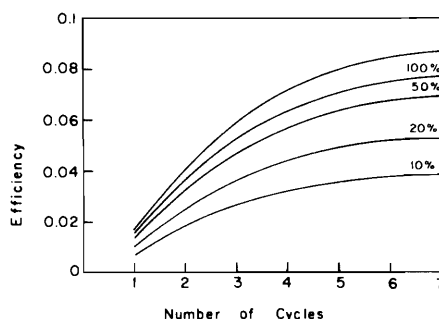


FIG. 8. Locomotory efficiency is plotted against the number of swimming cycles for the standard medusa. The top line corresponds to simulations neglecting energetic losses associated with bell deformation. Lines below that correspond to simulations including energy imparted to the bell with 100, 50, 20, and 10% efficiency of elastic energy storage.

the locomotor efficiency against the number of repeated swimming cycles for the standard medusa with a variety of elastic energy storage efficiencies. The top line of this figure corresponds to a calculation of locomotory efficiency with neglect of any energy imparted to the bell. Two features of the energetics of medusan locomotion are illustrated here: (i) A decrease in elastic energy storage gives rise to a decrease in the overall locomotory efficiency in a predictable way; (ii) Elastic energy storage during the first swimming cycle (escape swimming) is less important to locomotory efficiency than in continuous (repeated) swimming.

Timing of movements: behaviors

By varying the parameters of the standard medusa we may explore (i) the effect of contraction time on the initial acceleration of a medusa starting from rest and (ii) the effect of the relative phase duration on (a) the average acceleration during the first complete swimming cycle and (b) the average maximum velocity, the cost and the efficiency during normal swimming.

Figure 9a (solid line) shows that, not unexpectedly, higher accelerations are achieved with shorter contraction times. Note, however, that a shorter time to expel a given mass of water necessarily results in a greater energy expenditure (Fig. 9a, broken line): higher acceleration can only be achieved at the expense of increased energy input.

Figure 9b shows a potential constraint on the escape behavior of a swimming medusa. This figure is a plot of the cost/benefit ratio given by Eq. 22 which includes the energy (*E*) required to accelerate the standard medusa of average mass *m* at an average acceleration *a* through a distance *d*. This ratio, the escape index, ($\epsilon = E/m \cdot a \cdot d$) is minimized at a contraction time of 0.1 s. Below 0.1 s, the ratio increases exponentially and the "cost" associated with escape swimming progressively overwhelms the "benefits." For contraction times

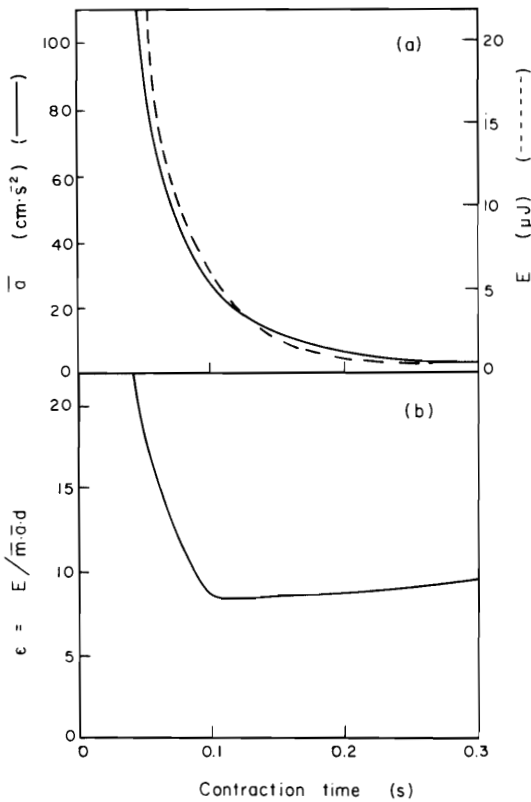


FIG. 9. (a) The average acceleration, $\bar{a}$, during the contraction phase (—) and the energy, E , required to produce that motion (---) are plotted against contraction phase duration for a hypothetical medusa 1 cm tall and 1 cm in radius. (b) The escape index ($\epsilon = E / m \cdot \bar{a} \cdot d$) during the contraction phase for the medusa is plotted against contraction time. An elastic energy storage efficiency of 100% was assumed in generating these functions. A decrease in the elastic energy storage efficiency does not affect the location of the minimum in panel b, but does yield a higher magnitude for the minimum.

greater than 0.1 s, there is only a slight increase in the ratio.

Figure 10 is a plot of the average acceleration of the standard medusa during the first complete swimming cycle after starting from rest for a variety of relative phase durations (see Eq. 1), considering only the acceleration for one complete cycle rather than the acceleration during the contraction phase alone. When the relaxation and contraction phase have equal durations, the net thrust for one complete swimming cycle is zero and the animal's average acceleration and velocity is zero. Only when the relaxation phase lasts longer than the contraction phase (relative phase duration > 1), can there be a net increase in the animal's velocity. A relative phase duration between 2.0 and 2.5 maximizes the average acceleration during this cycle.

This same value for relative phase duration maxi-

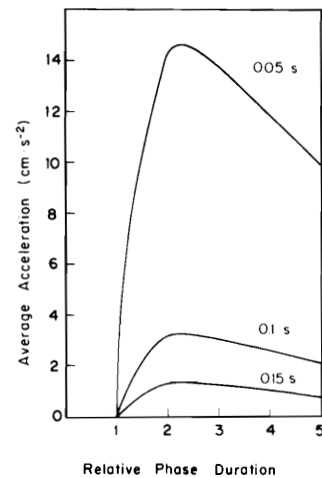


FIG. 10. The average acceleration during one complete swimming cycle is plotted against relative phase duration for a medusa 1 cm tall and 1 cm in radius starting from rest, exhibiting contraction times of 0.05, 0.1, and 0.15 s. A relative phase duration of about 2 maximizes the average acceleration regardless of the duration of the contraction phase.

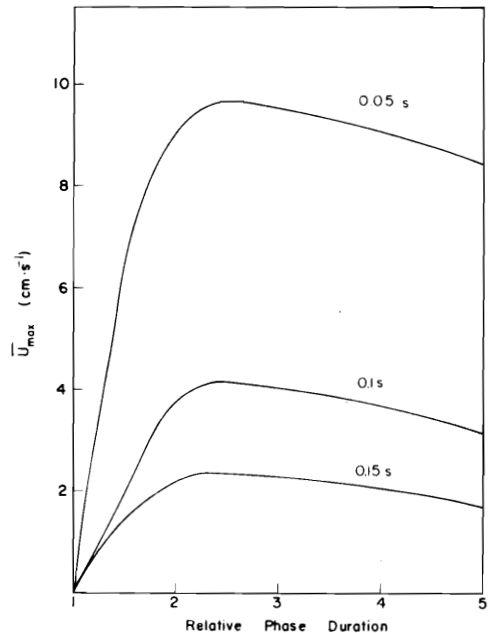


FIG. 11. Average velocity during normal swimming ($\bar{U}_{\text{max}}$) is plotted against relative phase duration for a medusa 1 cm tall and 1 cm in radius exhibiting contraction times of 0.05, 0.1, and 0.15 s. A relative phase duration of about 2 maximizes $\bar{U}_{\text{max}}$ regardless of the contraction phase duration.

mizes the average velocity during normal swimming ($\bar{U}_{\text{max}}$). Figure 11 is a plot of $\bar{U}_{\text{max}}$ against the relative phase duration for the standard medusa with a variety of contraction times. Notice that at any one contraction time $\bar{U}_{\text{max}}$ is maximized at a relative phase duration

between 2.0 and 2.5. As contraction time increases the magnitude of $U_{\max}$ decreases. Similarly, locomotor efficiency is maximized at the same value for the relative phase duration. The value of the maximum increases with decreasing contraction times. The relative phase duration which maximizes locomotory efficiency and maximum average velocity is independent of the efficiency with which the bell stores elastic energy.

Size and shape

Escape swimming

The escape index ($\epsilon = E/m \cdot a \cdot d$) for medusae of various shapes expelling 50% of their subumbrellar volume in 0.1 s is given in Fig. 12a as a function of the size (mass) of the animal. Recall that low values for this index are considered to be "better." Each shape is specified by the ratio of a medusa's bell height to its radius. Notice that for any shape escape performance is fairly independent of the mass of the animal. This result is somewhat surprising because the resistance to acceleration is directly proportional to the mass of the animal. It can be shown, however, that the thrust produced for a given contraction time and volume change will increase with the mass of the animal to the $\frac{5}{3}$ power. Hence increasing size will increase both propulsive and resistive forces and will, therefore, have little effect on escape performance. If the acceleration reaction is neglected for an animal of any one shape and size, the escape index will be considerably lower. For example, simulations for the escape index of the standard medusa predict ϵ to be 9.0 when the acceleration reaction is included; 3.0 when it is neglected. So, while the mass of an animal has little apparent effect on escape performance, the acceleration reaction is important.

Recall that the acceleration reaction depends on the shape as well as size of an animal. As Figs. 12a and 12b show, the shape of a medusa has a considerable effect on escape performance. More prolate medusae, by virtue of lower added-mass coefficients, exhibit a "better" escape (a lower escape index). Below a height/radius ratio of 1.0, the "cost" of escape increases exponentially relative to the "benefits", suggesting a shape constraint on the utility of escape swimming.

Normal swimming

Figure 13 (broken line, η) shows that for hemispherical medusae the efficiency of locomotion increases with body mass. Based on this simulation, we might expect selection in relation to swimming for large size. But the cost of locomotion (dashed line, C) also increases with body mass. Hence, the consequences of being large are both a high efficiency and a high cost. One may expect natural selection to favor those medusae which travel with the least cost and greatest efficiency. One possibility is for selection to favor medusae which travel with a maximum ratio of efficiency to cost. In fact, for a given

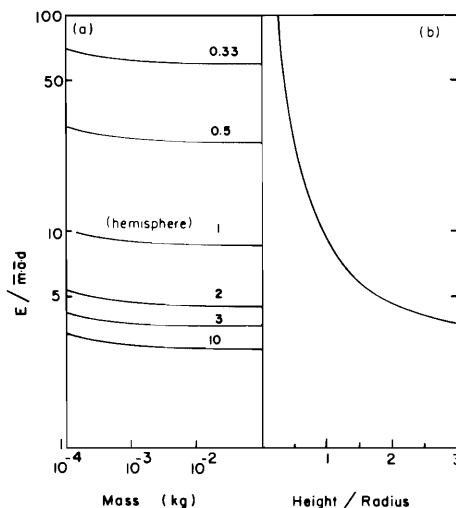


FIG. 12. (a) The escape index ($\epsilon = E/m \cdot a \cdot d$) is plotted against mass for medusae of a variety of shapes with a contraction phase lasting 0.1 s. Shape is specified by a bell height to radius ratio above each line. Notice that for any shape, the escape index is fairly independent of size. (b) The escape index is plotted against the bell height to radius ratio for a hypothetical medusa 100 g in mass with a contraction phase lasting 0.1 s.

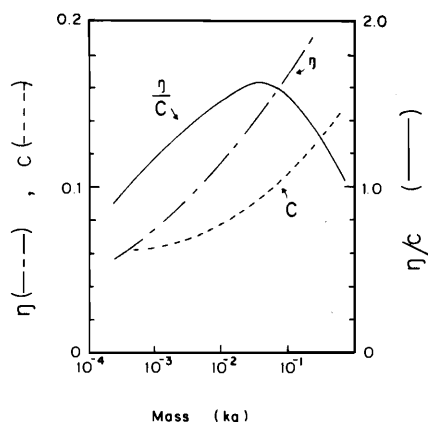


FIG. 13. Cost (C , ---), efficiency (η , —), and the efficiency to cost ratio (η/C , —) are plotted against body mass for hemispherical medusae with contraction phase durations of 0.1 s. The efficiency to cost ratio is maximized at a body mass of approximately 20 g.

shape and swimming behavior, this ratio is maximized at one particular size. For hemispherical medusae, the efficiency/cost ratio (η/C) is maximized at a mass of about 20 g, corresponding to a medusa 2 cm in bell height and radius.

The mass which maximizes the efficiency/cost ratio is independent of the shape of a medusa. Figure 14 is a plot of η/C against size for medusae of various shapes. Notice that the maximum value for η/C occurs at a mass

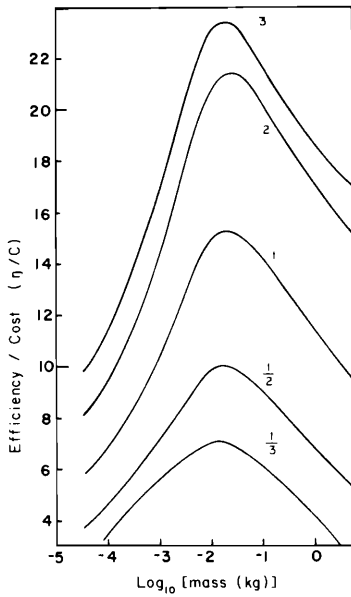


FIG. 14. The efficiency to cost ratio (η/C) is plotted against body mass for medusae of a variety of shapes with contraction phases lasting 0.1 s. Shape is specified above each line by a number equal to the ratio of bell height to radius. The mass which maximizes η/C is dependent of shape.

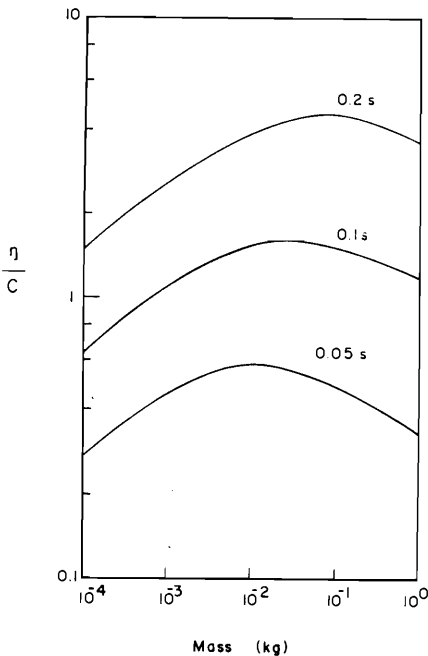


FIG. 15. The efficiency to cost ratio (η/C) is plotted against body mass for hemispherical medusae exhibiting a variety of contraction times (specified above each line). The mass which maximizes η/C increases with increasing contraction times.

TABLE 1. Comparison of theoretical predictions with existing kinematic data from Gladfelter (1973) for the hydrozoan *Liriope tetraphylla*. Mean values are reported; standard deviations were extracted from the author's data

Parameter	Measured	Theoretical
Bell height (cm)	1.5	1.0 (assumed)
Bell radius (cm)	1.0*	1.0 (assumed)
% volume change	47 ± 8.1	50 (assumed)
Maximum velocity (cm/s)	5.0 ± 1.1	5.2 (calculated)
Average velocity (cm/s)	2.9 ± 0.9	3.4 (calculated)
Average acceleration (cm/s ²)	$20.7 \pm 6.7^\dagger$	27.0 (calculated)

*Not reported explicitly. Values extracted from author's figures.

†Not reported explicitly. Values extracted from reported kinematic data.

of 20 g, regardless of the animal's shape. Also note that as medusae become more oblate, the magnitude of the optimum decreases and the curve becomes shallower. More oblate medusae may, therefore, be larger without a significant decrement in swimming performance (η/C).

Size and behavior

Swimming behavior has a considerable effect on the magnitude of η/C and the size at which that ratio is maximized. Figure 15 is a plot of η/C against body mass for medusae exhibiting various contraction times. This figure illustrates two general trends: (i) As contraction time increases, the mass of the medusa which gives rise to a maximum efficiency/cost ratio also increases and (ii) the longer the duration of the propulsive phase, the greater the magnitude of the efficiency/cost ratio for a medusa of any size.

Test of the model

In Table 1, I compare characteristics and the theoretical predictions for the kinematics of the standard medusa with the data of Gladfelter (1973) for the hydrozoan medusa, *Liriope tetraphylla*. This animal has approximately the same dimensions and swimming behavior as the standard medusa.

Discussion

Mechanics

Thrust

The thrust created by expulsion of water was considered to be equivalent to the thrust produced by a steady jet. But, medusae swim with repeated contractions of their bell and the jet, therefore, is produced periodically. Weihs (1977) and Seikman (1963) have shown that a pulsed jet can give rise to a greater average thrust force than a steady jet of equivalent mass flow rate. In a pulsed jet, an ejected mass of fluid rolls into a toroidal vortex ring which moves away from its source. A continuously pulsating jet, therefore, produces a row of these vortex rings each with some radius b and spaced a units

apart. When the spacing between rings is less than three ring radii ($a < 3b$), the average propulsive force will be higher than that generated by an equivalent steady jet. For example, when the ring spacing is equal to the ring radius ($a = b$) thrust is augmented by about 50% relative to a steady jet (Weihs 1977). Is thrust augmentation likely to occur in medusae?

A 1-cm tall hemispherical medusa may produce vortex rings approximately 0.5 cm in radius. Each ring will move away from the medusa at a velocity about equal to the jet velocity. So, for a contraction phase of 0.1 s, I calculate the ring velocity to be about 15 cm/s. Assuming a relaxation phase duration of 0.2 s, rings will be produced every 0.3 s for a continuously pulsing medusa. Thus, before a second vortex ring is produced, the first one will have traveled about 5 cm and the distance between rings will be about 10 ring radii. We may, therefore, neglect thrust augmentation due to a pulsed jet.

Resistance

The model shows that the acceleration reaction is the largest instantaneous force acting on a swimming medusa. But the acceleration reaction may not be very important as a force averaged over a complete swimming cycle: in normal (continuous) swimming, the time averaged acceleration of a medusa is zero and the animal, therefore, experiences no net acceleration reaction. Even though the acceleration reaction is important as an instantaneous force, its average contribution to the total fluid force acting on a medusa is zero. In escape swimming, however, the acceleration reaction exceeds drag as an instantaneous or time averaged force.

Clearly, mechanical analyses of other unsteady motions among animals must consider this force. Hence, the thrust produced by animals which swim with rowing or oscillating appendages should be determined, in part, by the acceleration reaction. Indeed, as Lighthill (1975) demonstrated, the acceleration reaction is essentially the only force giving rise to thrust for large undulating (anguilliform) swimmers.

Applications of the acceleration reaction to medusan locomotion and unsteady motions in general are subject to an important limitation. As any object accelerates in a fluid, it generates vorticity which diffuses away from that object by the action of viscosity and creates vortices in the flow about the object. Vortices give rise to time-varying flow patterns and thus to added-mass coefficients which vary in time (Batchelor 1967; Birkhoff 1960). Birkhoff (1960) suggests that analytical solutions for the added-mass coefficient are applicable for the first few diameters of travel. This condition is satisfied for the standard medusa. The model predicts its velocity to be about 5 cm/s. During a contraction phase of 0.1 s, the medusa would only travel about 0.5 cm, a

distance that is a small fraction of the bell diameter. Thus, application of the acceleration reaction to medusan locomotion is acceptable.

Escape swimming

Only a few medusae utilize their jet propulsion mechanism to escape predation. Most medusae involute the margin of their bell, "crumpling" in response to some mechanical stimulus (Donaldson *et al.* 1980; Mackie and Singla 1975). Here, then, we have two different behavioral responses to mechanical stimulation. Are there constraints on the behavior and morphology of medusae that give rise to this behavioral dichotomy? A partial answer to this question is suggested in the energetics of escape swimming.

One appropriate "strategy" for increasing the likelihood of escape is for a medusa to accelerate as rapidly as possible and to place as great a distance as possible between itself and its predator. Maximizing the product of acceleration and distance maximizes the likelihood of escape from predation (if a simple product determines success). One way to achieve a high acceleration-distance product is for a medusa to contract its bell as rapidly as possible (see Fig. 9). But, in view of the fact that the total energy required to produce a jet increases exponentially with decreasing contraction times, high accelerations are only achieved at the expense of high energy outputs.

Simulations show that the escape index ($\epsilon = E/m \cdot a \cdot d$) is minimized at a contraction phase duration of 0.1 s regardless of a medusa's size or shape. This minimum represents an energetic constraint on escape swimming. For contraction phase durations greater than 0.1 s, the cost of escape is low but so are the benefits (benefit = $m \cdot a \cdot d$). Contraction times less than 0.1 s give rise to very high acceleration-distance products but the costs become increasingly high. We may, then, expect medusan escape jetting to last on the order of 0.1 s. Indeed, those medusae which do exhibit jet propulsion as an escape behavior contract their bells in about 0.1 s. The hydrozoan, *Aglantha digitalis*, contracts its bell in 0.07 to 0.1 s during escape (Donaldson *et al.* 1980). The range of contraction phase durations for *Gonionemus vertens* during escape swimming is 0.1 to 0.14 s (T. L. Daniel, in preparation). In contrast, *Polyorchis* exhibits the "crumple" behavior in response to mechanical stimuli and contracts its bell in times greater than 0.2 s (Gladfelter 1973). And other animals which use jet propulsion for escaping predation show similar behaviors. Squid, ranging between 3 and 100 g in size, produce powerful jets lasting 0.085 to 0.12 s (Packard 1969). A dragonfly larva expels water through its anus in about 0.1 s for its escape response (Hughes 1958).

The model illustrates a potential constraint on the

shape as well as behavior of a jet propelling animal. Figures 12a and 12b show that the shape, not size, of an animal is an important determinant of its escape performance. For bell height/radius ratios less than 1.0 the costs of escape overwhelm the "benefits." Hence, if energy is limited, we may expect medusae demonstrating escape responses to be prolate. And, indeed, existing data support this hypothesis: *A. digitalis* and *G. vertens* are generally prolate with height/radius ratio as great as 3 (Donaldson *et al.* 1980; T. L. Daniel, in preparation); squid and dragonfly larvae are quite prolate with length/radius ratios greater than 3.0 (Hughes 1958; Packard 1969; Ward and Wainwright 1972).

Normal swimming

Hydrozoan medusae show diverse swimming behaviors associated with feeding (Mills 1981). Some species swim upwards in the water column for a metre or so then sink slowly, capturing small organisms in their tentacles. And many species undergo daily vertical migrations, feeding at the surface by day or night (Mackie 1974). For all these animals, translation through their environment is necessary for feeding. And, according to "optimal foraging theory" (for a review see Pyke *et al.* 1977), those animals minimizing the energy required to feed should, in an evolutionary sense, be favored.

One aspect of medusan swimming that determines the energetics of locomotion is the relative phase duration. For example, when the relative phase duration is 1.0, the thrust generated during the contraction phase is equal in magnitude but opposite in sign to the thrust generated during the relaxation phase, and the average velocity of a medusa swimming in this fashion will be zero (assuming no changes in the velar aperture between phases). Alternatively, when the duration of the relaxation phase is much greater than that of the contraction phase, the average velocity of the animal will be quite low since most of the swimming cycle is spent decelerating to a stop. The simulation shows that a relative phase duration of about 2.0 yields a maximum efficiency and swimming velocity and a minimum cost of locomotion. The relative phase duration which gives rise to this minimum and these maxima is independent of the size and shape of a medusa and the duration of the contraction phase.

Existing data show that the relative phase duration for medusae is highly variable. Donaldson *et al.* (1981) report a relative phase duration between 3 and 4 for *A. digitale*. Gladfelter's (1972, 1973) data show relative phase durations between 1.3 and 2.5 depending on the species. It appears, then, that these values bracket the predicted "best" relative phase duration. And, again, the hypothesis of energetic constraints on behavioral aspects of medusan locomotion is suggested by existing data.

Size and behavior

One expects natural selection to favor those animals which move about with the greatest efficiency and the least cost if (i) moving through the environment is important to the survival of an animal and (ii) energy is in some way limited. Cost and efficiency, however, do not reflect similar aspects of animal locomotion. Efficiency is a measure of the rate at which an animal does work on its environment relative to the rate at which it consumes energy. Cost, on the other hand, is a measure of the rate at which an animal consumes energy relative to the rate at which it does work in moving itself. I extend this argument to consider the ratio of efficiency to cost as a reasonable measure of the locomotor performance of an animal. Selection might favor maximization of this ratio.

The efficiency/cost ratio is useful in illustrating constraints on the size and swimming behavior of medusae. For example, the simulation results show that medusae with contraction phases of 0.1 s should have a volume of approximately 20 cm³ to maximize this ratio. Larger medusae should exhibit slower contraction times to achieve maximum efficiency to cost ratios. Hence, for any one contraction phase duration there exists a unique size which maximizes the efficiency/cost ratio. Does swimming behavior scale with size so as to maintain a maximum efficiency/cost ratio? Existing data provides equivocal evidence for such a trend. Figure 16 is a plot of contraction phase duration against the mass at which the ratio is maximized. Included in this figure are data

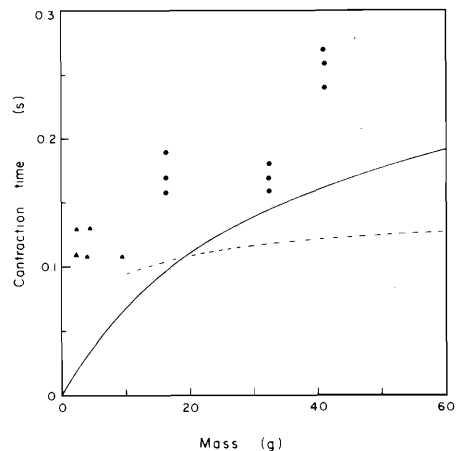


FIG. 16. Contraction time is plotted against body size (mass in grams) using data from T. L. Daniel (in preparation) for *Gonionemus vertens* (▲); Gladfelter (1973) for *Liriope tetraphylla*, *Stomatoca atra*, and *Phialidium hemisphericum* (●); and Spencer and Satterlie (1981) for *Polyorchis penicillatus* (broken line obtained from regression of author's data). The solid line in this figure represents the predicted mass which maximizes η/C for any contraction time.

from several sources (see figure legend) for contraction time as a function of body size for various hydrozoan medusae. It appears that the efficiency/cost ratio is a reasonable measure of swimming performance and is useful in predicting relationships between size and swimming behavior.

Limitations

Constraints on size, shape, and swimming behavior have been illuminated by the model. Underlying these constraints is the concept that animals must operate optimally. Thus an animal's behavior, size, or shape are judged by whether they give rise to maximal performance in escape or normal swimming. Certainly there are medusae which perform swimming movements "poorly." For example, simulation results show that medusae should be prolate to maximize efficiency and minimize the cost of locomotion. But not all medusae are prolate. So, while the theoretical model predicts that prolate animals are more "successful," we find many oblate medusae which translate poorly and exhibit no escape swimming response. Here, then, is the main limitation of the approach used in this paper: it views only energetic aspects of an animal's biology related to swimming. Natural selection may not optimize any one process.

Consider, for example, how the shape of a medusa determines the energetics of its locomotion. The model predicts that prolate medusae swim with a lower cost and a higher efficiency than more oblate medusae regardless of size. It is possible, however, that translation is not the primary object of swimming movements. The main function of swimming movements of a large, oblate, and "poor" swimmer may be to generate feeding currents. Prolate medusae translate effectively and may encounter prey much larger than themselves; more oblate animals translate poorly and may encounter only those prey small enough to be trapped in the entrained flow. Viewing the implications of shape only in terms of locomotion may be misleading. Shape may also be a mechanism for resource partitioning among medusae. Size and shape relate to aspects of an animal's biology other than locomotion and feeding: larger medusae may be more visible to predators than small ones; small medusae may exploit habitats not available to larger medusae; and gonad volume, hence reproductive output, may increase with body size.

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Appendix: Notation

Symbol	Quantity	Units
a	Acceleration	cm/s^2
A_v	Area of velar aperture	cm^2
C	Cost of locomotion	—
C_d	Drag coefficient	—
d	Distance	cm
D	Drag	N
E	Energy	J
E_f	Energy to fill bell	J
E_t	Total energy imparted to bell	J
F	Force to overcome acceleration reaction and inertia of animal	N
G	Acceleration reaction	N
h	Bell height	cm
l	Characteristic length for Re	cm
m	Mass of animal	g
P_i	Power input	W
P_o	Power output	W
$P_{i,f}$	Power to fill bell	W
$P_{i,t}$	Power to produce thrust	W
$P_{o,a}$	Power to overcome acceleration reaction	W
$P_{o,d}$	Power to overcome drag	W
r	Radius of animal	cm
R	Relative contribution of acceleration reaction to total resistive force	—

Appendix: Notation (*Concluded*)

Symbol	Quantity	Units
Re	Reynolds number	—
S	Area projected in direction of motion	cm ²
S ₀	Initial area projected in direction of motion	cm ²
T	Thrust	N
t _c	Duration of contraction phase	s
t _r	Duration of relaxation phase	s
u	Instantaneous velocity	cm/s
u _e	Velocity of ejected water	cm/s
U _{max}	Maximum velocity	cm/s
V	Volume	cm ³
V ₀	Initial volume	cm ³
W	Weight of animal	N
α	Added-mass coefficient	—
δ	Relative phase duration	—
Δ	Damping factor	—
ε	Escape index	—
η	Efficiency of locomotion	—
ν	Kinematic viscosity of water	cm ² /s
ρ	Density of water	g/cm ³
τ	Arbitrary time interval	s